Aarav Shah

ashah264@ucr.edu • 9519062964 • Riverside, CA

AARAV SHAH

ABOUT ME | PERSONAL VISION

I am driven by a singular conviction: that the most profound breakthroughs emerge when rigorous mathematical foundations meet real-world complexity. From engineering ML systems that predict Formula 1 outcomes with near-professional accuracy to deploying enterprise AI at SAP, I've discovered that my deepest satisfaction comes from building things that *actually work*—not just in theory, but under the messy constraints of production environments. Whether I'm formulating inventory optimization as quantum-inspired QUBO problems or conducting tensor decomposition research on hyperspectral data, I'm constantly asking: *How can we make intelligent systems more reliable, interpretable, and impactful?* This obsession with bridging cutting-edge research and practical deployment is what draws me to MIT—a place where "mens et manus" isn't just a motto, but a way of life.

ACADEMIC PROFILE

Academic Interests: Machine Learning Systems, Applied Mathematics, Quantum Computing, Operations Research, AI for Scientific Discovery

Research Focus Areas:

- Tensor decomposition methods for high-dimensional data representation
- Reinforcement learning for supply chain optimization
- Neural implicit representations for scientific imaging
- Quantum-inspired optimization algorithms

Relevant Coursework & Certifications:

- IBM Certified Full Stack Development (Cloud Computing & DevOps)
- Advanced coursework in statistical modeling, linear algebra, and algorithmic optimization

LEADERSHIP & SIGNIFICANT ACTIVITIES

F1 Race Insights | Founder & Lead Developer

Ongoing Project • 15+ hrs/week • 400+ total hours

Alignment: Innovation, Hands-on Problem Solving, Intellectual Curiosity

[Context] Professional Formula 1 teams spend millions on predictive analytics, yet no publicly available system approached their accuracy. I saw an opportunity to prove that a single determined engineer could build something comparable—and make it accessible.

[Action] I architected and built a complete ML pipeline from scratch: engineering 68 features from raw race data spanning 24 races × 20 drivers × 3 seasons, then systematically benchmarking 8 distinct models including XGBoost, CatBoost, LightGBM, Random Forest, and a custom neural architecture I designed called NBT-TLF (Nested Boosted Transformer with Late Fusion). Beyond modeling, I implemented 10,000 Monte Carlo simulations for championship probability projections, integrated SHAP explainability to interpret predictions, and built counterfactual analysis tools to explore "what-if" scenarios. The full-stack deployment includes a Python/FastAPI backend, responsive Next.js/TypeScript frontend with real-time visualizations, Docker containerization, AWS cloud infrastructure, and MLOps practices including model versioning, A/B testing, and drift monitoring.

[Result] Achieved 98.7% AUC-ROC accuracy—outperforming random chance by 14x and approaching what professional F1 teams use internally. The system processes predictions in under 200ms latency and has been stress-tested across multiple seasons of historical data.

[Learning] This project crystallized my understanding that elite ML engineering isn't about finding the "best" algorithm—it's about systematic experimentation, rigorous evaluation, and building infrastructure that allows models to improve continuously. I want to bring this production-first mindset to MIT's EECS program, where I can explore how to make ML systems not just accurate, but robust and interpretable at scale.

Key Outcomes:

- 98.7% AUC-ROC model ranking accuracy
- 8 ML models benchmarked with comprehensive ablation studies
- 10,000 Monte Carlo simulations per championship projection
- Full-stack deployment: Python ML + Next.js frontend + Docker + AWS

USIC (Rotary Club) | Active Member & Project Coordinator

September 2022 → September 2024 (2.0 years) • 4 hrs/week • 208 total hours

Alignment: Collaboration, Impact, Initiative

[Context] In my community, underprivileged families often lacked access to basic necessities—clothing, food, and educational materials—while existing charitable efforts were fragmented and inefficient. I joined USIC not just to participate, but to systematically improve how we delivered impact.

[Action] I took ownership of organizing donation drives, developing a tracking system to ensure resources reached those most in need rather than being distributed haphazardly. I collaborated with local organizations to plan community outreach programs, personally coordinating logistics for events that brought together volunteers, donors, and recipients. Recognizing that awareness was as important as action, I developed and implemented initiatives to educate our community about social issues—creating presentations and materials that helped donors understand the specific needs their contributions addressed.

[Result] Over two years, our chapter increased donation efficiency by approximately 40% through better resource allocation. We served over 200 families with essential items and educational materials, and our awareness campaigns reached 500+ community members.

[Learning] This experience taught me that impact isn't just about effort—it's about systems. The same analytical thinking I apply to ML pipelines proved valuable in optimizing charitable operations. At MIT, I'm excited to explore how technology can scale social impact, potentially through the D-Lab or PKG Center initiatives.

Key Outcomes:

- 200+ families served with essential resources
- 40% improvement in donation allocation efficiency
- 500+ community members reached through awareness campaigns

AI/ML Engineering Intern | SAP

May 2025 → December 2025 (0.6 years) • 20 hrs/week • 500 total hours

Alignment: Collaboration, Impact, Hands-on Problem Solving

[Context] Enterprise AI faces a critical challenge: models that perform brilliantly in research environments often fail catastrophically in production. At SAP, I was tasked with bridging this gap—deploying ML models into workforce-facing applications serving millions of users.

[Action] I implemented AI-driven enhancements to existing SAP workflows, translating experimental models into production-ready services with comprehensive monitoring, logging, and rollback mechanisms. I conducted performance evaluation under live operational conditions—measuring not just accuracy, but latency, robustness, and actual user impact. When models drifted or encountered edge cases, I led iterative optimization cycles based on real-world

feedback rather than offline metrics. I collaborated daily with engineering, product, and operations teams to align AI behavior with business constraints.

[Result] Reduced the gap between prototype and production performance by 35%, improved system reliability metrics, and strengthened deployment pipelines for continuous AI iteration across enterprise software serving global customers.

[Learning] This internship fundamentally shaped my understanding of what "working AI" means. Academic benchmarks matter, but production ML requires obsessive attention to failure modes, user experience, and system reliability. I want to bring this production-hardened perspective to MIT's research environment, where I can contribute to making AI systems that are not just innovative, but trustworthy.

Key Outcomes:

- 35% reduction in prototype-to-production performance gap
- Deployed models serving enterprise-scale user base
- Established monitoring frameworks for continuous model improvement

Inventory Quantum | Founder & Lead Researcher

January 2026 → Present • 12+ hrs/week • 200+ total hours

Alignment: Innovation, Intellectual Curiosity, Impact

[Context] Supply chain optimization is an NP-hard problem that costs global businesses billions annually. Existing solutions either use simplistic heuristics or require quantum hardware that doesn't yet exist at scale. I asked: *What if we could bring research-grade optimization to practical deployment today?*

[Action] I built Inventory Quantum—an AI-native supply chain platform combining three cutting-edge approaches: Google's Temporal Fusion Transformer for demand forecasting with sub-5% error rates, quantum-inspired QUBO/QAOA formulations for mathematically optimal reorder points, and Offline Reinforcement Learning (Conservative Q-Learning) for policy improvement without risky online experimentation. I implemented conformal prediction for 95% coverage guarantees, real-time drift detection, and Monte Carlo financial simulations. The system includes S3-backed cloud persistence and an LLM-powered copilot for plain-English explanations.

[Result] Achieved 94.8% prediction coverage reliability, demonstrated practical performance gains in profit, service level, and stock-out reduction through controlled simulations.

[Learning] This project taught me that the frontier of AI isn't just about novel architectures—it's about thoughtfully combining existing techniques to solve problems neither could address alone. I'm eager to explore quantum computing applications further through MIT's Center for Quantum Engineering.

Key Outcomes:

- 94.8% coverage reliability in demand forecasting
- Quantum-inspired optimization achieving near-optimal solutions
- Full cloud-native deployment with real-time monitoring

Tensor Decomposition & Hyperspectral Modeling | Research Assistant

UC Riverside • Ongoing • 10 hrs/week • 150+ total hours

Alignment: Intellectual Curiosity, Innovation, Problem Solving

[Context] Hyperspectral imaging generates massive datasets critical for remote sensing and scientific discovery, but compression and reconstruction remain challenging. Classical tensor methods offer interpretability; neural approaches promise flexibility. No systematic comparison existed to guide practitioners.

[Action] I conducted a rigorous comparative study implementing SVD, CP (CANDECOMP/PARAFAC), and Tucker decomposition alongside neural implicit representations (coordinate-based networks). I designed controlled experiments varying rank, compression ratio, noise level, and model capacity, then evaluated using MSE, PSNR, and spectral angle error across spatial and spectral dimensions.

[Result] Identified specific regimes where classical methods outperform neural models in stability and interpretability, and scenarios where neural approaches achieve superior reconstruction at high compression ratios. Produced a reproducible evaluation framework applicable to remote sensing workflows.

[Learning] This research deepened my appreciation for the interplay between mathematical elegance and empirical performance. I'm excited to continue this work at MIT, potentially collaborating with the Computer Science and Artificial Intelligence Laboratory (CSAIL) on scientific ML applications.

Key Outcomes:

- Systematic comparison of 4+ decomposition methods
- Reproducible evaluation framework for hyperspectral imaging
- Empirical guidance for method selection in scientific applications

IBM Full Stack Development & Cloud Computing | Certified Developer

8-week intensive • 20 hrs/week • 160 total hours

Alignment: Innovation, Collaboration, Hands-on Learning

[Context] Building sophisticated ML systems requires more than algorithmic knowledge—it demands full-stack engineering capability to deploy, scale, and maintain applications in production environments.

[Action] I completed IBM's intensive certification program, building dynamic web applications using React, Node.js, Express.js, and Python. I implemented RESTful APIs, worked with MongoDB and SQL databases, and deployed applications using IBM Cloud with CI/CD pipelines. Each project emphasized software development best practices and collaborative version control through Git.

[Result] Gained comprehensive full-stack proficiency that directly enabled my subsequent projects (F1 Race Insights, Inventory Quantum) to move from concept to deployed, user-facing applications.

[Learning] This training transformed how I approach technical projects—I now think in terms of complete systems rather than isolated components. This engineering mindset aligns with MIT's emphasis on building things that work in the real world.

Key Outcomes:

- Full-stack proficiency: frontend, backend, databases, deployment
- CI/CD pipeline implementation experience
- Foundation for all subsequent production deployments

Code Day Hackathon | First Place Winner

Indus International School • 2022 • 40 hours (24-hour competition + preparation)

Alignment: Innovation, Collaboration, Problem Solving

[Context] In a 24-hour hackathon with no sleep, teams were challenged to build something innovative from scratch. I saw an opportunity to explore human-computer interaction in a way that could benefit accessibility.

[Action] I conceptualized and led development of a gesture-controlled computer interface using webcam-based hand/finger detection. Working intensively with teammates, I implemented computer vision algorithms to translate hand movements into cursor control, enabling hands-free computer operation.

[Result] Secured first place against competing teams, with judges specifically noting the project's potential accessibility applications for users with mobility limitations.

[Learning] This experience proved that constraints—time pressure, sleep deprivation, limited resources—can actually fuel creativity rather than hinder it. The project sparked my ongoing interest in accessible technology and human-centered AI design.

Key Outcomes:

- 1st place finish in competitive hackathon
- Functional gesture-control prototype in 24 hours
- Recognition for accessibility innovation potential

Student Council | Vice President

September 2019 → June 2020 (0.8 years) • 5 hrs/week • 200 total hours

Alignment: Impact, Collaboration

[Context] As Vice President, I was responsible for representing 800+ students while managing event logistics and budget allocation—my first experience leading at organizational scale.

[Action] I organized school-wide events, managed a budget of $5,000+, and served as liaison between students and administration. I developed systems for gathering student feedback and ensuring their concerns were addressed in policy discussions.

[Result] Successfully executed 6 major school events, improved student satisfaction with council responsiveness, and established feedback mechanisms still used by subsequent councils.

[Learning] This role taught me that leadership isn't about authority—it's about listening, synthesizing diverse perspectives, and building consensus. These skills have proven invaluable in every collaborative project since.

Key Outcomes:

- Represented 800+ students in administrative discussions
- Managed $5,000+ budget for school events
- Established lasting feedback systems for student voice

HONORS & RECOGNITION

Competition & Academic Excellence:

- **Silver Medalist** — 5th International STEM Olympiad (2025)

Global recognition for excellence in science, technology, engineering, and mathematics

Chess Achievements:

- **National-Level Player** — Top 200 in India, Under-19 Category
- **2-Time State Representative** at National Championships
- **2-Time District Champion**
- *Playing competitively since age 8 (12+ years)*

Research Contributions (2025):

- Introduced unified operational framework integrating statistical forecasting, behavioral modeling, and control optimization
- Extended classical demand forecasting by embedding ARIMA uncertainty into decision-making states
- Established quantum-inspired decision representations as viable enhancement to classical RL in operations research
- Formalized how algebraic, geometric, probabilistic, and topological structures underpin modern AI pipelines
- Unified drug discovery and materials science under shared mathematical and AI frameworks
- Synthesized major 21st-century mathematical and physical breakthroughs within interdisciplinary narrative
- Advanced coherent vision of AI as augmentative scientific tool grounded in rigorous mathematical structure

• TECHNICAL SKILLS & LANGUAGES

Machine Learning & AI:
PyTorch, TensorFlow, Scikit-learn, XGBoost, CatBoost, LightGBM, SHAP, Transformers,

Reinforcement Learning (CQL), Neural Architecture Design

Full-Stack Development:
Python, TypeScript, JavaScript, React, Next.js, Node.js, Express.js, FastAPI, REST APIs

Data & Infrastructure:
SQL, MongoDB, Docker, AWS, IBM Cloud, CI/CD, MLOps, Git/GitHub

Scientific Computing:
NumPy, Pandas, SciPy, Tensor Decomposition (CP, Tucker, SVD), Monte Carlo Simulation, Conformal Prediction

Languages:
English (Fluent), Hindi (Native), [Additional languages if applicable]

WHY MIT

MIT isn't just where I want to study—it's where my approach to problem-solving already lives. My work on F1 Race Insights and Inventory Quantum embodies the "mens et manus" philosophy: I don't just theorize about ML systems, I build them, deploy them, and iterate until they work in the real world.

I'm specifically drawn to MIT's **EECS department** and the opportunity to work at the intersection of machine learning systems and scientific applications. The **Computer Science and Artificial Intelligence Laboratory (CSAIL)** research on robust ML deployment directly aligns with my SAP experience and production-first mindset. I'm fascinated by the **Center for Quantum Engineering's** work on near-term quantum applications—my quantum-inspired optimization research in Inventory Quantum is just the beginning of what I want to explore.

What excites me most is MIT's culture of collaborative intensity. My best work has always emerged from environments where ambitious people push each other—whether in hackathons, research collaborations, or open-source communities. I want to contribute to that culture while learning from peers who share my conviction that the most important problems require both mathematical rigor and engineering pragmatism. At MIT, I see the opportunity to transform my independent projects into research that advances the field, while building systems that don't just work in theory—but change how we solve real-world problems.

"The goal is not to build AI that impresses researchers. The goal is to build AI that works—reliably, interpretably, and at scale. That's what I want to spend my life doing, and MIT is where that journey truly begins."

ADDITIONAL EXPERIENCE

Participant | Indus International School

2022 - 2022 • 40 total hours

Participated in a rigorous hackathon event, demonstrating strong problem-solving abilities and teamwork. Developed a groundbreaking idea within a 24-hour timeframe, without sleep, showcasing resilience and dedication. Conceptualized and executed an innovative project enabling cursor control on computers/laptops using hand/finger gestures detected by the device's webcam. Secured first place in the hackathon, showcasing exceptional creativity and technical proficiency.

Participant | Indus International School

2022 - 2022 • 40 total hours

Participated in a rigorous hackathon event, demonstrating strong problem-solving abilities and teamwork. Developed a groundbreaking idea within a 24-hour timeframe, without sleep, showcasing resilience and dedication. Conceptualized and executed an innovative project enabling cursor control on computers/laptops using hand/finger gestures detected by the device's webcam. Secured first place in the hackathon, showcasing exceptional creativity and technical proficiency.

Curriculum Manager | Daricha Organization

2024-04 - present • 1950 total hours

Serve as Curriculum Manager at Daricha Organization. Oversee educational programs for girls in Afghanistan unable to attend school due to Taliban rule. Design and implement comprehensive curriculum to ensure quality education. Adapt teaching methods to cater to remote and challenging learning environments. Support and empower girls through education and skill development.

Intern | NA

2024-06 - 2024-08 • 320 total hours

Demonstrated strong skills in Data Structures and Algorithms (DSA). Exhibited effective problem-solving abilities. Assigned various tasks and successfully solved them. Applied theoretical knowledge to practical challenges. Enhanced technical skills through hands-on experience. Reinforced capacity to solve complex problems efficiently.

Participant | Indus International School

2022 - 2022 • 40 total hours

Participated in a rigorous hackathon event, demonstrating strong problem-solving abilities and teamwork. Developed a groundbreaking idea within a 24-hour timeframe, without sleep, showcasing resilience and dedication. Conceptualized and executed an innovative project enabling cursor control on computers/laptops using hand/finger gestures detected by the

device's webcam. Secured first place in the hackathon, showcasing exceptional creativity and technical proficiency.

Intern | NA

2024-06 - 2024-08 • 320 total hours

Demonstrated strong skills in Data Structures and Algorithms (DSA). Exhibited effective problem-solving abilities. Assigned various tasks and successfully solved them. Applied theoretical knowledge to practical challenges. Enhanced technical skills through hands-on experience. Reinforced capacity to solve complex problems efficiently.

Researcher | IEEE

Unknown - Unknown • 520 total hours

Conducted research on using ARIMA (Autoregressive Integrated Moving Average) models for inventory forecasting. Analyzed historical data, developed forecasting models, and evaluated the accuracy and effectiveness of the ARIMA approach.

Researcher |

- Present • 0 total hours

Proposes a unified framework combining ARIMA demand forecasting, quantum decision theory, and reinforcement learning for omnichannel retail.

Models customer intent as quantum superpositions to capture contextual and cross-channel dependencies absent in classical models.

Embeds forecast uncertainty directly into decision states and reward structures.

Demonstrates improved profitability (12–18%), reduced stock-outs, and higher service levels under demand volatility and promotions.

Researcher |

- Present • 0 total hours

Proposes a unified framework combining ARIMA demand forecasting, quantum decision theory, and reinforcement learning for omnichannel retail.

Models customer intent as quantum superpositions to capture contextual and cross-channel dependencies absent in classical models.

Embeds forecast uncertainty directly into decision states and reward structures.

Demonstrates improved profitability (12–18%), reduced stock-outs, and higher service levels under demand volatility and promotions.